Digital Watermarking

Digital watermarking

At a very general level digital watermarking is a way of hiding a message (called stegomessage or watermark) within a host (cover) signal.

By adopting a communication point of view the following more formal definition can be adopted: Watermarking is a mechanism to create a communication channel that is multiplexed into original content.

Tom Kalker "Considerations on watermarking security" IEEE International Workshop on Multimedia Signal Processing 2001, pp. 201-206, Oct. 2001.

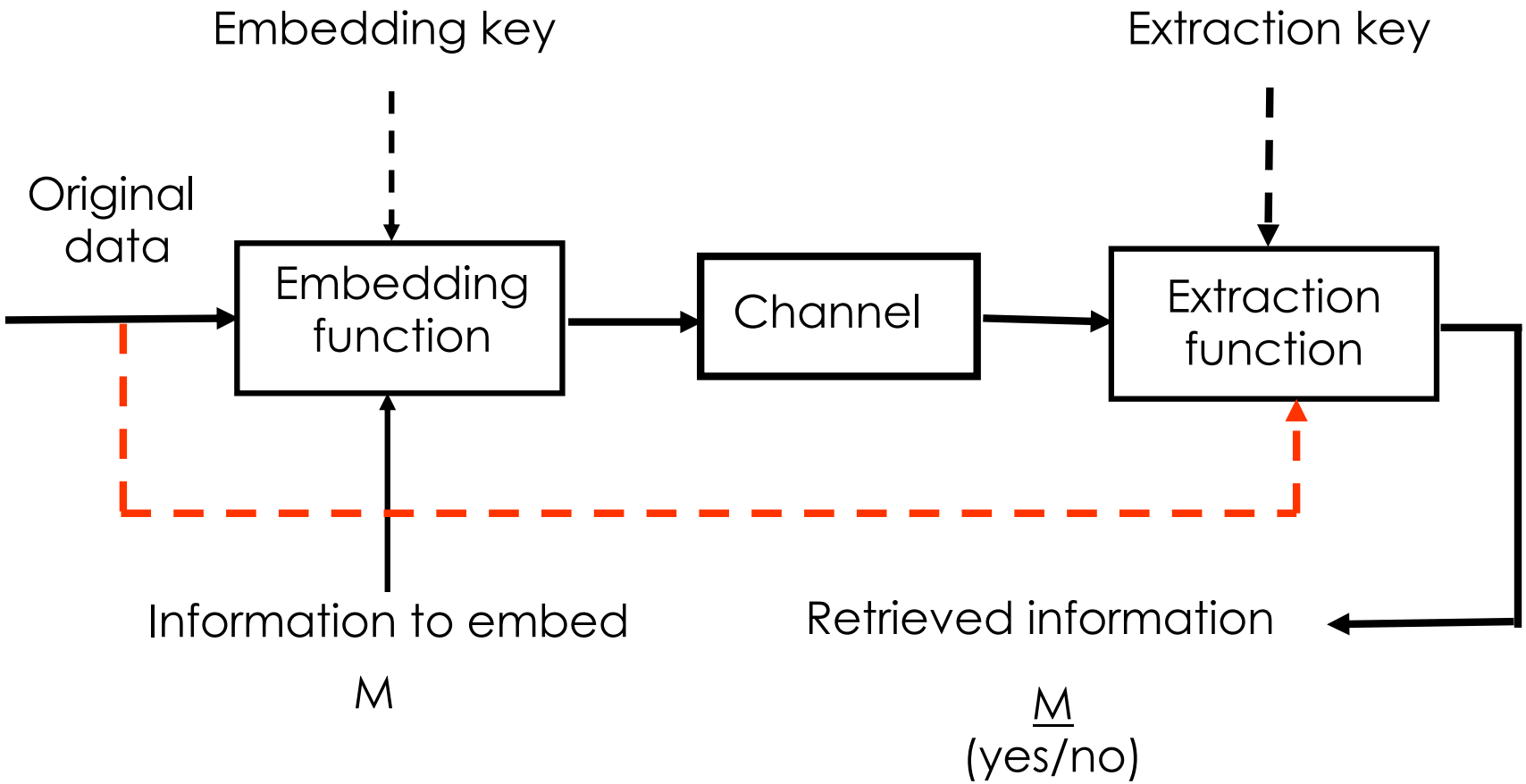
Digital watermarking

How can information be hidden in digital data?

By exploiting "perceptual headroom"

- Human perception is imperfect
- Make modification to the original data without changing its perceptual quality
- Modifications can be detected via signal processing

Theoretical model 1



Requirements

- Imperceptibility
- Robustness - The embedded data should survive any signal processing operation the host signal goes through and preserve its fidelity
- Capacity - Maximize data embedding payload

Requirements

- Security - Kerckhoff's principle states that one should assume that the method used to encrypt the data is known to an unauthorized party and that the security must lie in the choice of a key

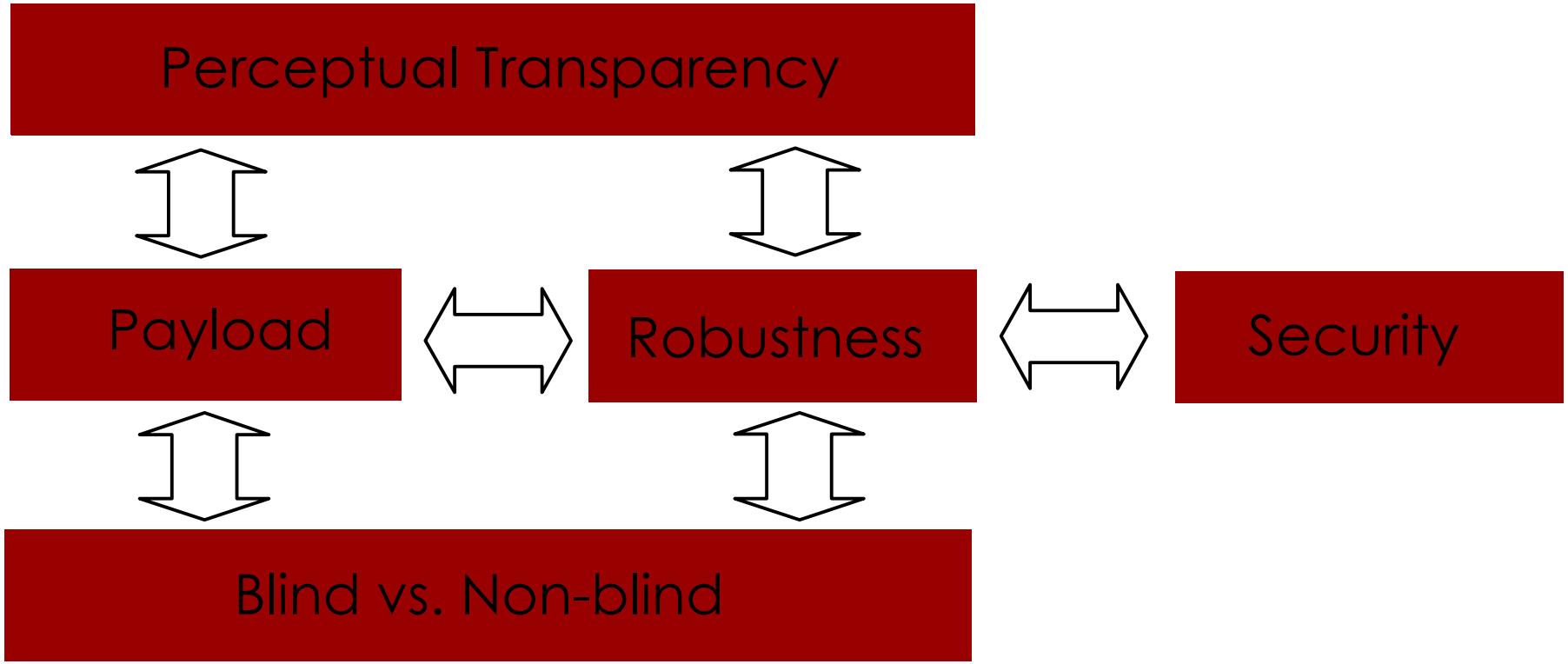
Active Attacks

- Hacker has access to some data and attempts to extract information about secret data

Passive Attacks

- Hacker tries to get information about secret data without playing with data (eavesdropping)

Requirements



Requirements

- Robust watermarking is required in copyright protection and ownership identification scenarios
- A fragile watermark is needed for authentication and tamper proofing applications
- A statistically invisible watermark is mandatory for steganography
- A high payload is required for labelling applications and legacy systems

Classification

- Data hiding can be classified in different ways based on
- the relative importance of cover and message signals
- the nature of cover content
- the need for an unmodified cover signal for message signal extraction
  - blind
  - non-blind
- the type of attacks to remove hidden messages

Classification

- Blind techniques
  - the watermark is recovered without resorting to the original non-marked image or any information derived from it

- Non-blind techniques
  - the original image is needed to read the watermark
  - robustness is more easily achieved
  - often the application scenario does not allow the decoder to access the original non-marked image

Classification

- Decoding process
  - detectable (1-bit, 0-bit) watermark
    - it is only possible to decide whether a given watermark is embedded in the image
    - privateness is automatically achieved
  - readable watermark (multibit watermarking)
    - the bits hidden in the image can be read without knowing them in advance

Classification

- Symmetric vs. asymmetric schemes
  - in order to improve security in a public detection scenario asymmetric schemes have been proposed
  - the detection key is different than (a subset of) the embedding key
  - detecting the watermark should not help removing it

Classification

- Spatial domain watermarking - Watermark embedded by directly modifying the pixel values

- Transform domain watermarking - Watermark embedded in the transform domain e.g., DCT, DFT, wavelet by modifying the coefficients of global or block transform

Classification

- Spatial domain
  - Simplicity
  - More effective exploitation of HVS characteristics
  - Syncronization problems
- Transform domain
  - Higher robustness
  - Intrinsic resistance to some geometric manipulations
  - Lack of localization
  - High computational burder

Classification

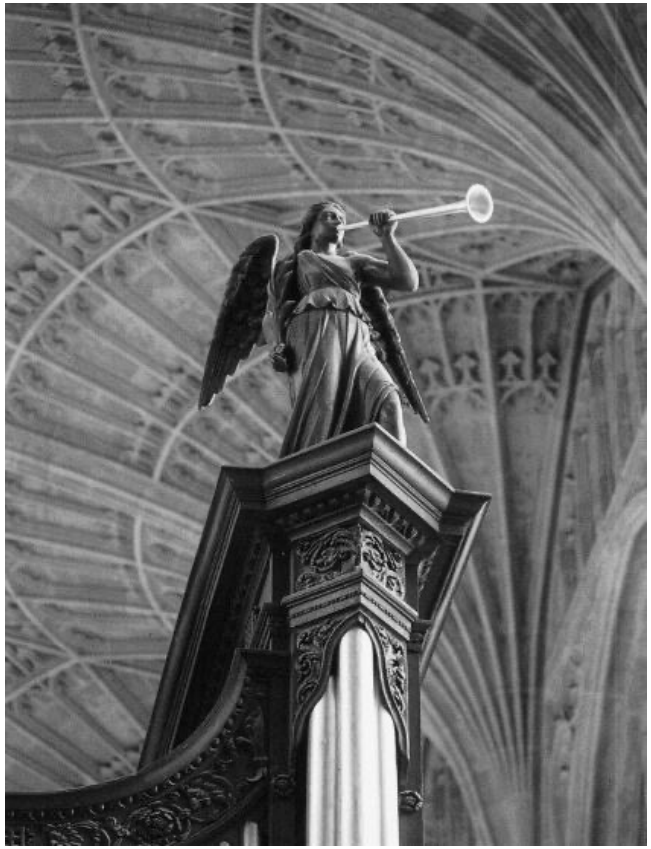
- Hybrid techniques
  - Advantages of frequency watermarking without losing spatial localization

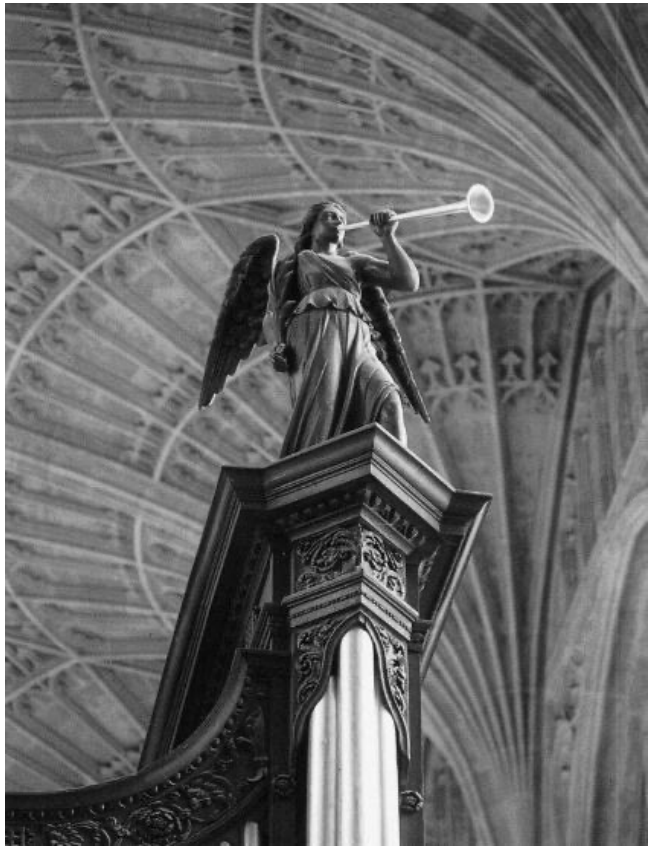
LSB replacement

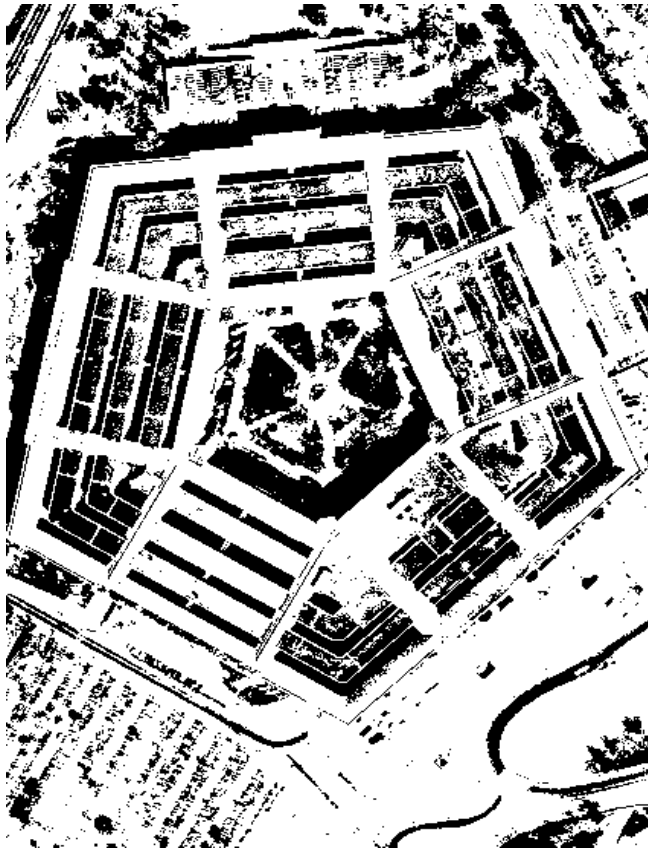
Results from LSB substitution: the watermarked image shows little not noticeable degradation, while the large watermark can be recovered perfectly.

LSB replacement

LSB replacement







Replace LSB with Pentagon's MSB

LSB replacement

……	
11	1011
10	1010
9	1001
8	1000
7	0111
6	0110
5	0101
4	0100
3	0011
2	0010
1	0001

how much distortion by embedding?
how much resilience to minor changes?

LSB replacement

Replace 6 LSBs with Pentagon's 6 MSBs

LSB replacement

8 bits / pixel

4 bits / pixel

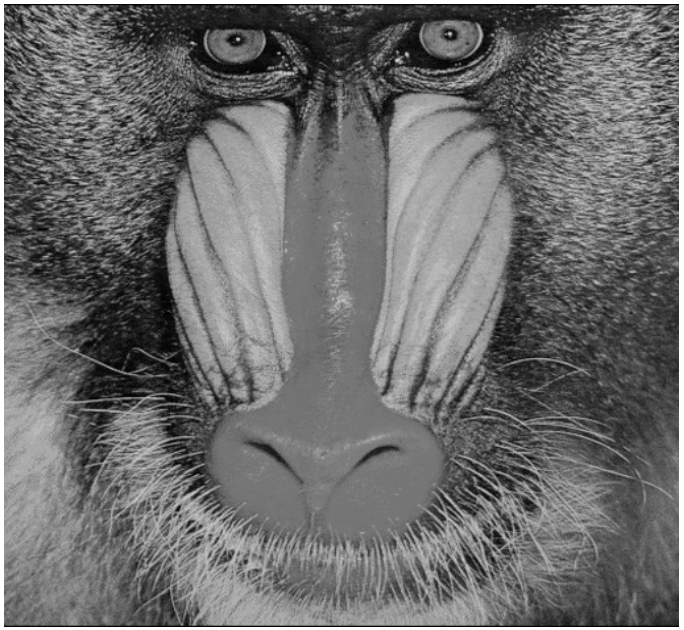
2 bits / pixel

Human eyes distinguish about 50 gray levels
=> 5~6 bits/pixel

LSB replacement

Original Image

Watermark



LSB replacement

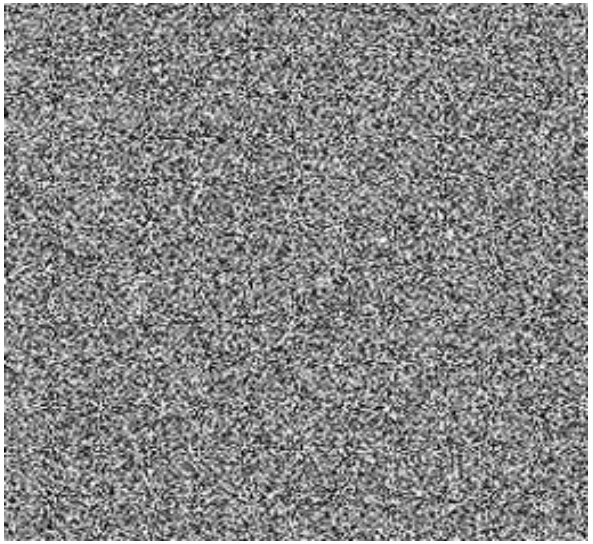
4 LSBs Watermarked

7 LSBs Watermarked

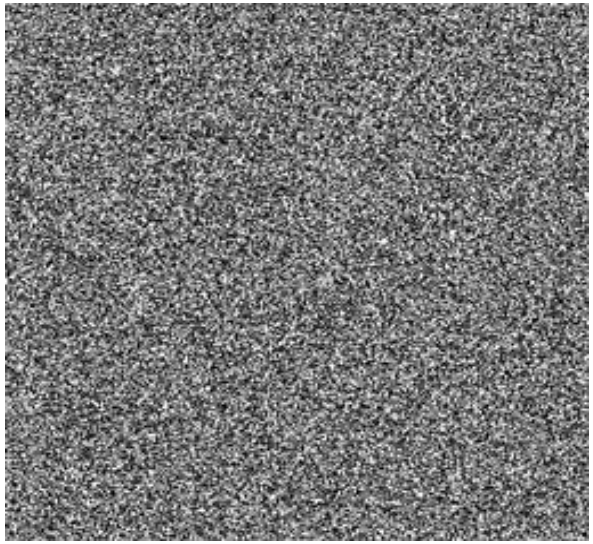
LSB replacement

Although the watermark was recovered perfectly in the ideal case, the addition of any amount of noise, or compression of the image using JPEG fully destroys the embedded watermark, leaving nothing but noise. Even worse, the watermark can be removed with no perceivable change to the watermarked image. The message capacity of LSB embedding however is quite good, a 1:1 correlation with the size of the image.

LSB replacement



Recovered watermark after addition of 1% Gaussian noise



Recovered watermark after JPEG compression with quality 95

LSB replacement

How is it possible to achieve robustness?

Robustness

Set of attacks

- Signal processing
  - Enhancement, sharpening, blurring, linear/nonlinear filtering
- Compression
  - Robustness against JPEG compression is mandatory
  - JPEG 2000
- Geometric manipulations
  - Resizing, cropping, translation, rotation, flip

Robustness

For a quantitative measure of robustness, a given attack, or combined attack, is made at different levels, the higher the level the watermark survives the higher the robustness.

Use of watermark-breaking tools

- Stirmark
- Certimark

Robustness vs. security

- Robutness deals with non-malicious manipulations
- Security consider malicious attacks in a hostile environment
- In a security analysis it is assumed that the attacker knows the watermarking algorithm: hence ad hoc attacks can be conceived

Robustness vs. security

- Attacks to robustness are those whose target is to increase the probability of error of the data hiding channel

- Attacks to security are those aimed at gaining knowledge about the secrets of the system

Fernando Perez-Gonzalez et al. "First summary report on fundamentals" ECRYPT – European Network of Excellence in Cryptology, IST-2002-507932, D.WVL.1, 2005.